

CSE 717 — Information Security

Topic 12: Blockchain & Bitcoin — Architecture, Hashing/PoW, Bitcoin Mining

Complete Past Paper Solutions: 2020–2024

Source: Lc#11A (Blockchain), Lc#11B (Bitcoin Mining)

2/5 years confirmed (2021, 2024), 3–8.75 marks. NEW dedicated 2026 materials — expect strong emphasis.

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Quick Reference — Blockchain & Bitcoin (CHEAT SHEET)

Block structure (Lc#11A p2)

Every block has a header containing:

1. **Version** — version of the block validation rules.
2. **Previous Block Header Hash** — reference to this block's parent — **this is what chains the blocks together**.
3. **Merkle Root Hash** — a cryptographic hash of ALL transactions in this block (hash of hashes, combined pairwise up a tree).
4. **Time** — approximate creation time.
5. **nBits** — the current Proof-of-Work difficulty target.
6. **Nonce** ("**number used once**") — a random counter miners vary to find a valid block hash.

Below the header: the **coinbase transaction** (miner's reward) + the list of **transactions**.

How hashing makes the chain secure (Lc#11A p3–4)

- Each block's hash is a unique "fingerprint" of its contents. ANY change inside a block changes its hash.
- Block n stores the hash of block $n - 1$ as its **Previous Hash** field.
- If an attacker edits block 2's data, Block 2's hash changes $\Rightarrow$ Block 3's stored "Previous Hash" no longer matches Block 2's new hash $\Rightarrow$ the tamper is immediately detectable.
- To successfully tamper, an attacker would have to recompute the hashes of **every subsequent block** — and (in a real network) outrun the honest nodes' Proof-of-Work, which is computationally infeasible.

Proof of Work (PoW) (Lc#11A p4)

- A PoW is a **computational puzzle**: find a **nonce** such that $\text{hash}(\text{block header} \parallel \text{nonce})$ produces a hash below a target value (e.g. starts with a required number of leading zeros).
- This is hard to solve but **easy for others to verify** (just hash once and check).
- Miners repeatedly try different nonces until one works — this is "mining."
- **Why it secures the chain**: to alter a past block, an attacker must redo its PoW *and* the PoW of every block after it, faster than the rest of the network combined — effectively impossible once a block has several confirmations.

Merkle Tree — data-integrity check (Lc#11A p2)

- Each transaction is hashed individually (leaf nodes).
- Pairs of hashes are concatenated and hashed again, repeatedly, up to a single **Merkle root**.
- The Merkle root is stored in the block header.
- **Integrity check**: if ANY transaction in the block changes, the Merkle root changes $\Rightarrow$ mismatch with the header $\Rightarrow$ tampering detected *without* re-checking every transaction individually.

Distributed P2P Network & consensus (Lc#11A p5–6)

1. Every node in the P2P network holds a full (or partial) copy of the chain.
2. To add a block: (1) one node will need to update all chains on the network; (2) redo the proof-of-work for each block; (3) take control of more than 50% of the peer-to-peer network.
3. **Consensus** = nodes agree on what blocks are valid; **decentralization** means no single point of failure.

4 pillars of blockchain security (Lc#11A p7): Resilience (no single point of failure), Time-stamping, Reliability (certified transactions), Unchangeable transactions (immutability).

Types of blockchain (Lc#11A p8–9)

- **Public**: fully decentralized, anyone can join/read/write (e.g. Bitcoin, Ethereum).
- **Private**: access restricted, controlled by one organization (e.g. Hyperledger, R3 Corda).
- **Hybrid**: combines public security/transparency with private confidentiality (e.g. Dragonchain).

Blockchain vs. Shared Database (Lc#11A p10)

Parameter	Blockchain	Shared Database
Operations	Insert only	Create/Read/Update/Delete
Replication	Full replication on every peer	Master-slave / multi-master
Consensus	Most peers agree on outcome	2-phase commit / Paxos
Validation	Global rules enforced network-wide	Only local integrity constraints
Disintermediation	Allowed	Not allowed
Confidentiality	Fully confidential	Not totally confidential

Bitcoin essentials (Lc#11B)

- **Bitcoin $\neq$ Blockchain.** Bitcoin is a consensus network + digital currency; blockchain is the underlying technology that records its transactions.
- **Mining** = competing to add the next block by solving the PoW puzzle; the winner broadcasts the block and collects the **block reward** + transaction fees.
- **Mining difficulty** self-adjusts so that, on average, one block is found every ~ 10 minutes (Bitcoin), regardless of total network hash power.
- **Mining pools** let many miners combine hash power and split rewards proportionally — reduces reward variance.
- **Wallet** = software holding your private/public key pair; controls access to funds recorded on-chain (the coins themselves never "move" off-chain).
- **Bitcoin exchange** = where BTC is bought/sold for fiat or other crypto (e.g. Binance, Coinbase).

2024 Q3(b) — Blockchain Security via Hashing + Proof of Work [3 marks]

Question — 2024 Q3(b) [3]

A company maintains a blockchain-based distributed ledger for its digital transactions. Each new block contains the hash of the previous block, forming a chain. To add a block, participants must solve a Proof of Work puzzle, making tampering computationally difficult. Now, explain how blockchain ensures security, specifically in terms of hashing and Proof of Work.

Answer

(1) Security via hashing (chain-linking):

- Each block's header stores the **cryptographic hash of the previous block**, creating a tamper-evident chain.
- A block's own hash is computed over all of its contents (header fields + Merkle

root of its transactions). Any change to even a single transaction changes the Merkle root, which changes the block's hash.

- If an attacker modifies block k , block k 's hash changes, so the "previous hash" field stored in block $k + 1$ no longer matches — the chain is immediately broken/detectable.
- To make the forged chain look valid again, the attacker must recompute the hashes (and headers) of **every block from k to the latest block**.

(2) Security via Proof of Work:

- Each block's hash must satisfy a difficulty target (e.g. a required number of leading zero bits); finding such a hash requires brute-forcing a **nonce** — computationally expensive but cheap to verify.
- Recomputing one block's PoW takes significant time/energy; recomputing the PoW for *all* subsequent blocks (to cover up a tamper) is exponentially harder.
- Meanwhile, honest miners keep extending the real chain, so the attacker's forged chain falls further behind and is rejected by the network's "longest valid chain" rule.

Conclusion: hashing makes any tampering *detectable* (chain of fingerprints), and Proof of Work makes tampering *computationally infeasible to hide* (every block after the tampered one must be re-mined faster than the honest network) — together they give the ledger its immutability and integrity guarantees.

2021 Section B Q3 — Blockchain/Cryptography Relation, Bitcoin, I/O Script Validation [8.75 marks]

Question — 2021 Section B Q3(a) [2.75 + 2]

- (i) How is the blockchain related with cryptography? [2.75]
- (ii) What do you understand about bitcoin? [2]

Answer (i) — Blockchain $\leftrightarrow$ Cryptography [2.75]

Blockchain relies on cryptography at every layer:

- **Cryptographic hash functions** link blocks together (each block stores the hash of the previous block, forming the chain) and build the **Merkle tree** that summarizes all transactions in a block — this gives **integrity and tamper-evidence**.
- **Public-key cryptography (asymmetric keys)**: each participant has a key pair. Transactions are **digitally signed** with the sender's private key and verified with their public key — this gives **authentication** (proof of ownership) and **non-repudiation**.
- **Proof of Work** is itself a cryptographic-hashing puzzle (find a nonce so the hash meets a difficulty target) — this gives **consensus/security** against tampering.

$\Rightarrow$ In short: hashing secures the chain's structure and data integrity; public-key

cryptography (digital signatures) secures ownership and authorization of transactions.

Answer (ii) — What is Bitcoin? [2]

- Bitcoin is a **decentralized digital currency** / **peer-to-peer electronic cash system** with no central bank or single administrator.
- Transactions are verified by network nodes via cryptography (digital signatures) and recorded on a public distributed ledger called a **blockchain**.
- New bitcoins are created (and transactions confirmed) through **mining** — nodes competing to solve a Proof-of-Work puzzle.
- **Note:** Bitcoin (the currency/network) is not the same as blockchain (the underlying technology) — bitcoin uses blockchain, but not all blockchains are bitcoin.

Question — 2021 Section B Q3(b) [4]

Discuss the process of input-output scripts for signature validation for bitcoin.

Answer — Input/Output Scripts & Signature Validation [4]

Every bitcoin transaction consists of **inputs** (references to coins being spent) and **outputs** (new coins being created/assigned to recipients). Spending is controlled by a tiny stack-based scripting system:

1. **Output script** (“**locking script**” / **scriptPubKey**): attached to each transaction output, it specifies the **condition** that must be satisfied to spend that output — typically “a valid signature matching a given public key/address.”
2. **Input script** (“**unlocking script**” / **scriptSig**): when the recipient later wants to spend that output, they provide an input script containing their **digital signature** (and public key).
3. **Validation process:**
 - (a) The node concatenates/executes the **input script** followed by the referenced **output script**.
 - (b) The script engine checks: does the supplied public key hash match the one in the output script? Does the supplied signature verify correctly against the transaction data using that public key?
 - (c) If the combined script evaluates to **TRUE** (signature valid + conditions met), the output is successfully “unlocked” and can be spent; the new transaction’s outputs become locked with new conditions for the next owner.
4. This mechanism proves the spender **owns the private key** corresponding to the coin’s address **without revealing the private key itself** — the core of bitcoin’s ownership/authorization model.

2020, 2022, 2023 — No Blockchain/Bitcoin question found

2020/2022/2023 papers

No Blockchain or Bitcoin question was identified in the 2020, 2022, or 2023 past papers (verified against the consolidated past-papers PDF). **Note:** an earlier tracker entry claimed a 12-mark "2022 Q8" Merkle-tree/ETH-vs-BTC question — this was checked against the actual 2022 page (which contains only Q1–Q5) and does *not* exist; it has been removed. Blockchain & Bitcoin is confirmed in 2/5 years (2021, 2024), with brand-new dedicated 2026 lecture materials (Lc#11A/11B) suggesting it is likely to appear again.

Bonus Cheat-Sheet — Likely New Question Types from Lc#11A/11B

If a fresh Q8-style question appears in 2026 (insurance, not yet seen in past papers)

- **Draw/explain the Merkle tree** for N transactions: pairwise-hash leaves up to a single root stored in the block header; explain how it gives $O(1)$ -ish tamper detection.
- **ETH vs Bitcoin:** Bitcoin = purely a digital-currency/payment network (PoW, ~ 10 min blocks); Ethereum = programmable blockchain supporting **smart contracts** (self-executing code), its currency is **ether (ETH)**, used to pay "gas" fees for computation, not just transfers.
- **Genesis block:** the very first block in a chain, has no predecessor (previous hash = 0000...).
- **Public vs Private vs Hybrid blockchain** — table above (permission model + examples).
- **Blockchain vs shared database** — table above (operations/replication/consensus/validation/confidentiality).
- **Limitations of blockchain (Lc#11A p12):** higher cost than centralized DB, slower transaction throughput, smaller ledger size limit, risk of 51% attack, irreversibility (no "undo").
- **Bitcoin mining mechanics:** block reward halving (every $\sim 210,000$ blocks ≈ 4 years), self-adjusting difficulty to keep ~ 10 min/block, mining pools to reduce reward variance.

Exam Strategy Summary

High-Yield Checklist for Blockchain & Bitcoin (2/5 years confirmed, 3–8.75 marks)

1. **Hashing + PoW = the core security story.** Memorize the 2024 answer pattern: chain-of-hashes makes tampering *detectable*, PoW makes tampering *computationally infeasible to hide* (must re-mine all subsequent blocks faster than the honest network).
2. **Blockchain $\leftrightarrow$ cryptography (2021):** hash functions $\rightarrow$ chain integrity + Merkle tree; public-key crypto $\rightarrow$ digital signatures for owner-

ship/authentication.

3. **Bitcoin definition (2021):** decentralized P2P digital currency, no central authority, mining-based consensus, NOT the same thing as "blockchain."
4. **Input/output scripts (2021):** output script = locking condition (spend requires valid signature); input script = unlocking proof (signature + pubkey); validation = combined script executes to TRUE.
5. **Bonus prep (new Lc#11A/11B material, not yet in past papers):** Merkle tree diagram, block-header fields (6: version/prev-hash/Merkle root/time/nBits/nonce), public vs private vs hybrid chains, blockchain vs shared DB table, ETH vs BTC, blockchain limitations.
6. **If asked to draw something:** default to the 3-block hash-chain diagram (Block $n - 1$ hash $\rightarrow$ Block n 's "previous hash" field) — it's the single most reusable diagram across all sub-questions.